# TONY CHEN

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## EDUCATION

**University of California, Los Angeles** Expected 2030

*B.S., Electrical and Computer Engineering*

**Shady Side Academy, Pittsburgh, PA** 2022 – 2026

- GPA 4.07/4.00. National Merit Semifinalist (Pennsylvania). ACT 36 composite; SAT 1560.
- AP Calculus BC (5), AP Computer Science A (5), AP Physics C: Mechanics (4), AP Physics C: E&M (4); coursework in linear algebra, differential equations, and calculus-based physics.

## EXPERIENCE

**NASA, Embedded Systems Intern (paid)** Summer 2025

- Developed an autonomous mapping and placement pipeline in ROS2 on a rover platform.
- Built a LIDAR detection system to map an unknown area and generate a spatial model of it.
- Applied OpenCV and AprilTag detection to identify and place areas within the newly mapped space.
- Demo: youtu.be/HZcPi-EScxw (mapping), youtu.be/xC2AR4yw_AI (placement).

## TECHNICAL SKILLS

**Programming:** Python, Java, ROS2, OpenCV

**CAD / CAM:** Autodesk Fusion 360 for full-assembly modeling, part design, and CAM toolpaths

**Fabrication:** FDM 3D printing (plastic parts), CNC milling (metallic parts), mechanical assembly

**Electronics:** Motor drive, electronic speed controllers, stepper motor control, LiPo power systems

## ENGINEERING PROJECTS

**Automated Tracking Telescope** 2025 – 2026

- Locates and tracks deep-sky objects using two stepper motors on independent axes.
- Camera attachment for long-exposure astrophotography, plus swappable eyepieces for variable magnification.
- Shutter system for brightness control; target magnification 200×, enough to resolve Jovian bands and Saturn's rings.

**Custom Remote-Controlled Race Car** 2024 – 2025

- 3D-printed chassis with a dual-motor drivetrain, independent electronic speed controllers, and a shared 5200 mAh 11.1 V LiPo pack.
- Ackermann steering geometry so the inner and outer wheels track at different optimal angles, which tightens the turning radius and keeps the tires from scrubbing in corners.
- 15° caster angle to generate self-aligning torque, improving straight-line stability at speed and steering return after turns.

**Science Olympiad Electric Vehicle** 2024 – 2025

- Arduino-controlled vehicle that runs a track as fast as possible and stops at a target point, scored on combined time and stopping distance. The 7–10 m target distance is announced only after impound, so the vehicle had to be recalibrated to an unknown distance and run twice inside an 8-minute window.

## LEADERSHIP AND ACTIVITIES

**Leader, Robotics Club and FTC Team at Shady Side Academy** 2023 – 2026

- Sole club leader: recruiting, team organization, fundraising, and direction of design, build, programming, and competition.
- Modeled the complete robot chassis from scratch in Fusion 360 using goBILDA components, adding a formal CAD phase the team had never had before, and wrote the control code in Java.
- Drove recruiting to ~80 signups from 600 students; raised ~$600 via crowdfunding and a bake sale to close the equipment funding gap.

**Computer Science Departmental Associate at Shady Side Academy** 2025 – 2026

- One of five students selected to represent the CS department at school events, tutor peers, and promote computer science programs.

**Writing Center Scribe at Shady Side Academy** 2025 – 2026

- Admitted as 1 of 20 applicants to the Theory and Process of Writing course and selected to serve as a writing center scribe.

**Science Olympiad, Pennsylvania** 2024 – 2026

- 3rd place in Bungee (Penn State Invitational) and Dynamic Planet (Solon Invitational), February 2025; also competed in Material Science, Electric Vehicle, and Optics.

**Music and Athletics** 2014 – 2026

- First violin, Pittsburgh Youth Philharmonic Orchestra (2019–2026); ABRSM Piano Grade 8 and Theory Grade 5, both with Distinction; JV soccer, basketball, and track & field.